Quantitative Methods I

Week 3 - The Derivative

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Motivation

Imagine you are approached by a bookseller,

- the bookseller's profits W are a function of the price p,

$$W(p) = -20p^2 + 1200p - 10000$$

- the bookseller is currently charging 25 euro per book

Questions:

- Do you advise the bookseller to increase prices?
- Does this make profits go up or down, and by how much?

To answer this question, we need a tool that measures the instantaneous rate of change. That tool is the derivative.

Motivation - cont'd

Loosely speaking, the derivative

- quantifies the rate of change of a function at a certain point
- as the ratio of the instantaneous change in the dependent variable to that of the independent variable

Via the derivative, we can determine

- marginal profits, i.e., how profits react to small changes in sales volume
- marginal revenue, marginal cost, marginal utility, the marginal propensity to consume

Average Rate of Change (or Difference Quotient)

The average rate of change of the function $f(x)$ over the interval $[a, a + \Delta x]$ equals

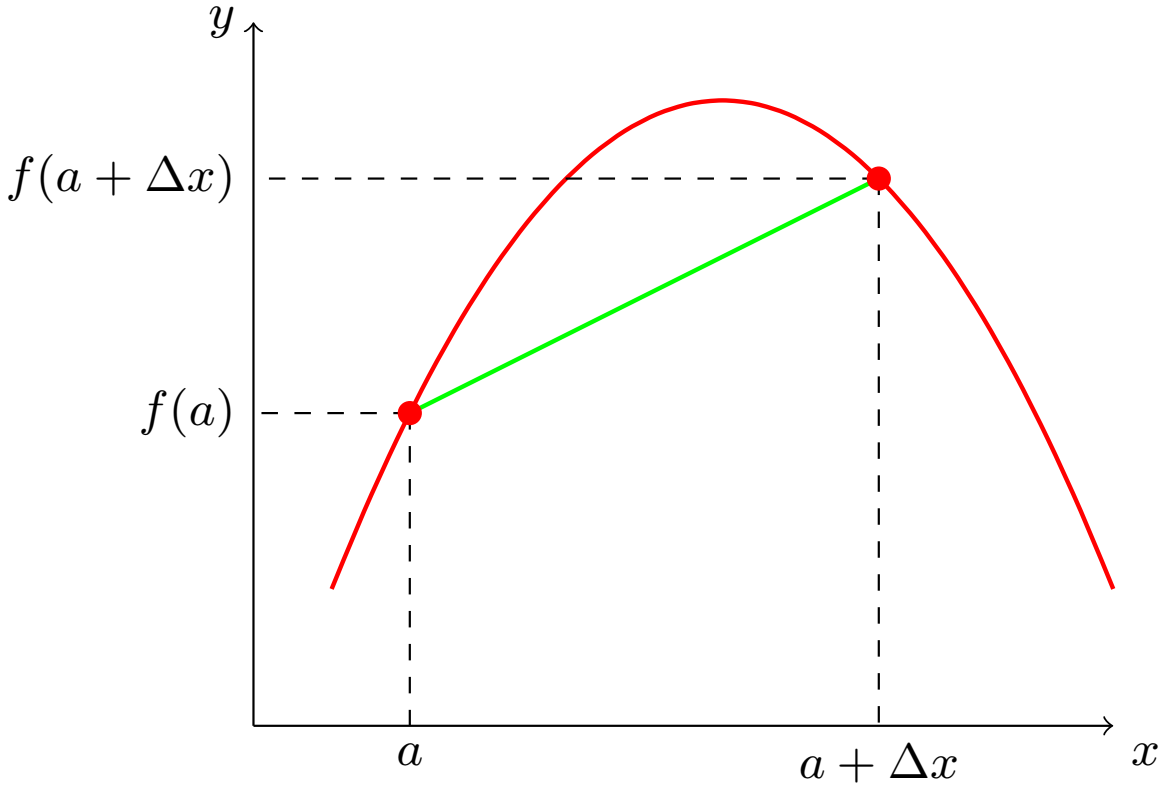
$$\frac{\Delta y}{\Delta x} = \frac{f(a + \Delta x) - f(a)}{\Delta x}$$

This fraction is also called the difference quotient of the function f corresponding with the change Δx

the average rate of change of the function f over the interval $[a, a + \Delta x]$
is the slope of the line through the endpoints of the graph of f on this interval.

Difference Quotient - graphically

$$\frac{\Delta y}{\Delta x} = \frac{f(a+\Delta x) - f(a)}{\Delta x} = \frac{f(a+\Delta x) - f(a)}{(a+\Delta x) - a}$$



Example

Recall our bookseller's profits $W(p)$ depend on the price p of a book,

$$W(p) = -20p^2 + 1200p - 10000$$

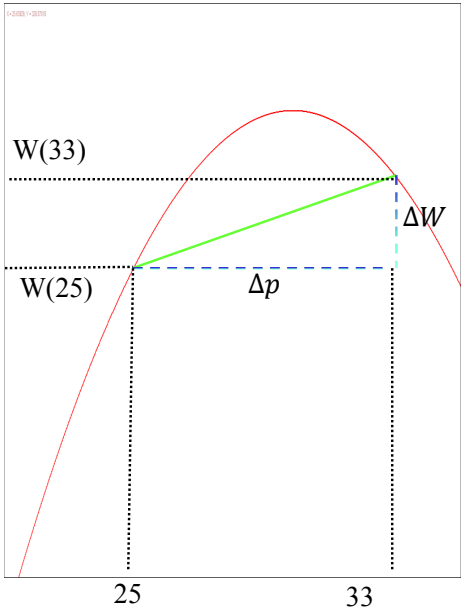
Q: What is the average rate of change in profit as p increases from 25 to 33?

Profit at respective prices,

$$W(25) = -20 \cdot (25)^2 + 1200 \cdot 25 - 10000 = 7500$$
$$W(33) = -20 \cdot (33)^2 + 1200 \cdot 33 - 10000 = 7820$$

Average rate of change,

$$\frac{\Delta W}{\Delta p} = \frac{W(33) - W(25)}{33 - 25} = \frac{7820 - 7500}{8} = \frac{320}{8} = 40$$



Rate of Change (or Derivative)

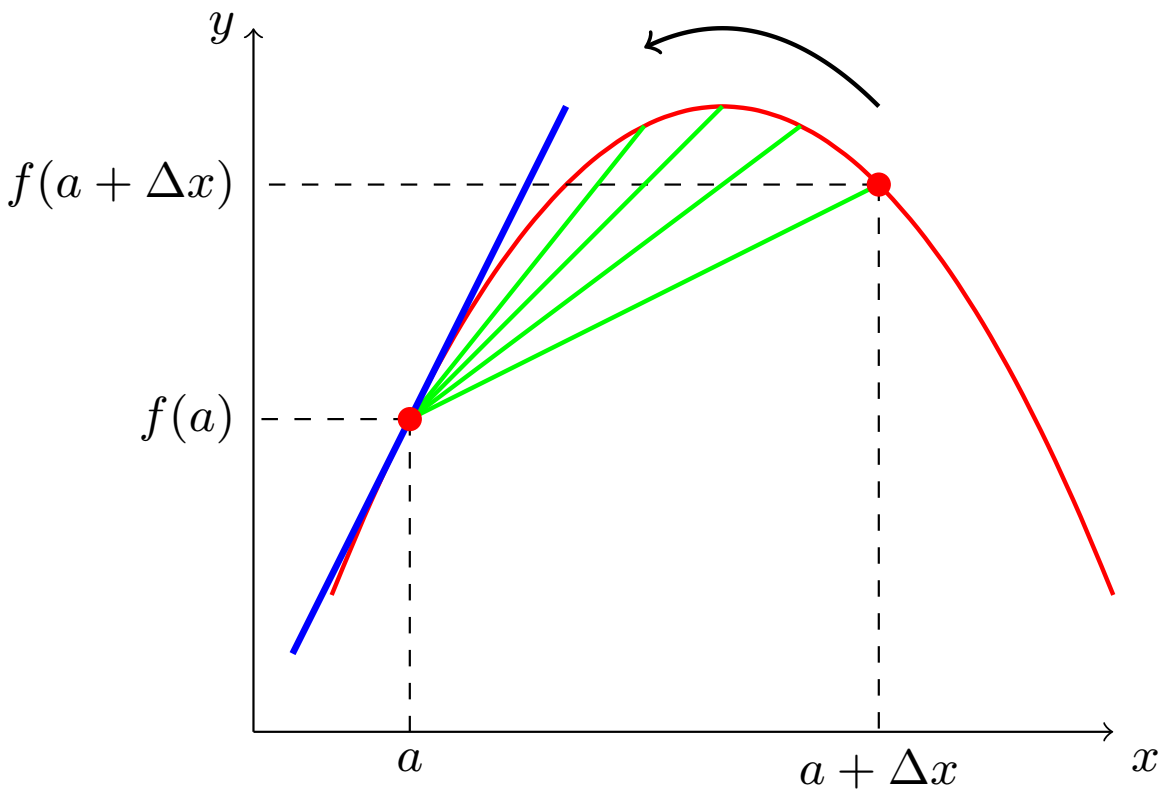
The rate of change of the function f at $x = a$ equals

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(a + \Delta x) - f(a)}{\Delta x}$$

This number will be denoted by $f'(a)$ and is usually referred to as the derivate of the function f at a

Derivative - graphically

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(a+\Delta x) - f(a)}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(a+\Delta x) - f(a)}{(a+\Delta x) - a}$$



Example

Recall that the bookseller's profit $W(p)$ was given as

$$W(p) = -20p^2 + 1200p - 10000$$

Q: What is the marginal profit of increasing the price from $p = 25$?

$$\begin{aligned}
\frac{\Delta W}{\Delta p} &= \frac{W(25 + \Delta p) - W(25)}{\Delta p} \\
&= \frac{(-20 \cdot (25 + \Delta p)^2 + 1200 \cdot (25 + \Delta p) - 10000) - (7500)}{\Delta p} \\
&= \frac{-1000 \cdot \Delta p - 20 \cdot (\Delta p)^2 + 1200 \cdot \Delta p}{\Delta p} = 200 - 20 \cdot \Delta p
\end{aligned}$$

Therefore,

$$\lim_{\Delta p \to 0} \frac{\Delta W}{\Delta p} = 200$$

Sum and Simple Product Rule

THE SUM RULE

If the derivatives f' and g' exist, then

$$[f(x) + g(x)]' = f'(x) + g'(x)$$

THE SIMPLE PRODUCT RULE

If the derivative f' exists and c is some constant, then

$$[cf(x)]' = cf'(x)$$

Computing Derivatives

For power functions, we have

$$f(x) = x^n \qquad \text{has derivative} \qquad f'(x) = nx^{n-1}$$

As such,

- $f(x) = x^5, \qquad f'(x) = 5x^4$
- $f(x) = x^{\frac{4}{3}}, \qquad f'(x) = \frac{4}{3}x^{\frac{1}{3}}$
- $f(x) = \frac{1}{x^3} = x^{-3}, \qquad f'(x) = -3x^{-4}$
- $f(x) = \sqrt{x} = x^{\frac{1}{2}}, \qquad f'(x) = \frac{1}{2}x^{-\frac{1}{2}}$
- $p(x) = 3 + 2x - 5x^3 + 8x^4, \qquad p'(x) = 2 - 15x^2 + 24x^3$

Back to the Bookseller

Remember the "hard way" we calculated the marginal profit at p=25 using limits?

- Let us now try the "easy way"

The profit function was:

$$W(p) = -20p^2 + 1200p - 10000$$

Using Power and Sum Rules, the derivative is:

$$W'(p) = (-20 \cdot 2p^{2-1}) + (1200 \cdot 1p^{1-1} - 0 = -40p + 1200$$

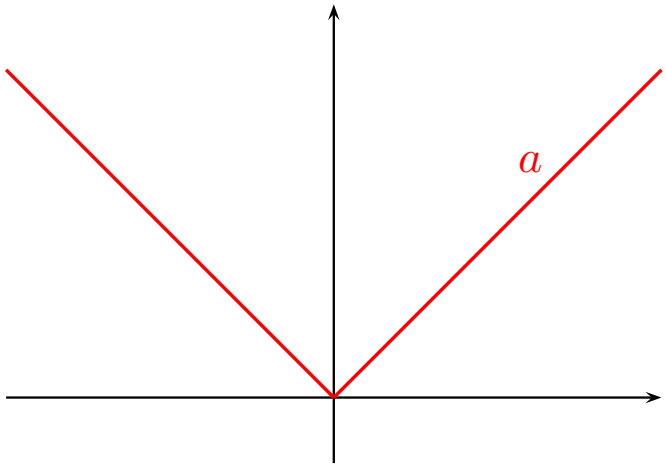
As such, the marginal profit at $p = 25$ is

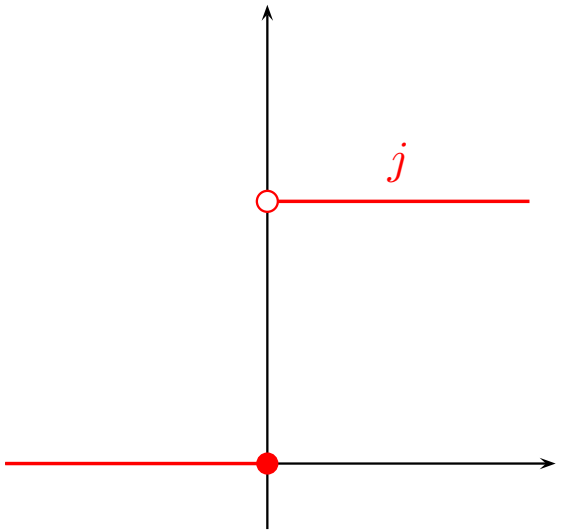
$$W'(25) = -40(25) + 1200 = -1000 + 1200 = 200$$

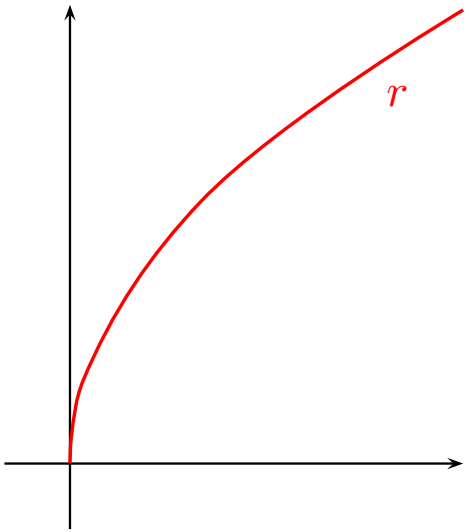
Note: Same result as before, but much faster!

Does the derivative always exist?

Consider the following situations (where the derivative does not exist at $x = 0$)







$$a(x) = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases} \qquad j(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases} \qquad r(x) = \sqrt{x}$$

Elasticity

If $y = f(x)$, where f is a positive function, then the elasticity at some value $x > 0$ is defined as

$$El_f(x) = f'(x) \cdot \frac{x}{f(x)}$$

The relative change $\frac{\Delta y}{y}$ in the variable y is approximately equal to $El_f(x)$ times the relative change $\frac{\Delta x}{x}$ in the variable x:

$$\frac{\Delta y}{y} \approx El_f(x) \frac{\Delta x}{x}$$

The elasticity translates a small relative change in x into an approximation for the resulting relative change in y.

Example

Bookseller still hasn't had enough,

$$W(p) = -20p^2 + 1200p - 10000$$
$$W'(p) = -40p + 1200$$

Q: What is the elasticity at $p = 25$?

From before, we had $W(25) = 7500$ and $W'(25) = 200$.

$$El_W(25) = W'(25) \cdot \frac{25}{W(25)} = 200 \cdot \frac{25}{7500} = 0.667$$

Therefore, at a price of 25 euros, a price increase of 1% leads to an increase of approximately 0.667% in the bookseller's profit.